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EFFECT OF REGRESSION TO THE MEAN IN THE PRESENCE OF WITHIN-SUBJECT VARIABILITY

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SUMMARY

Regression to the mean arises often in statistical applications where the units chosen for study relate to some observed characteristic in the extreme of its distribution. Gardner and Heady attribute the effect of regression to the mean to measurement errors. They assume the model $Y_i = U + e_i$, where U is a fixed within-subject component and e_i is the random measurement error. They suggest several replicate measurements to reduce the regression effect under the assumption that the measurement errors e_i are independent within subjects. While measurement errors play an important role in regression to the mean, one should not overlook within-subject variation. In this paper, we consider a model to estimate the regression effect in the presence of correlated within-subject effects as well as independent measurement errors.

INTRODUCTION

Sir Francis Galton[1] observed the tendency for tall parents to produce tall offspring but who on average are shorter than their parents. An analogous observation was made for offspring of short parents. He concluded that 'Each peculiarity in a man is shared by his kinsmen, but *on the average* to a less degree'. The linear equation used to model a trait measured on a parent and that same trait measured on an offspring gave rise to the general problem of regression analysis. The 'regression towards mediocrity' problem considered by Galton has come to be known as regression to the mean. Das and Mulder[2] suggested the terminology 'regression to the mode'; however, Senn[3] gave examples to suggest that indeed regression to the mean is more appropriate.

Regression to the mean occurs in a wide class of applications and is the subject of considerable discussion in the literature. Some clinical trials, for example, involve screening patients by observation of some trait and then choosing as eligible for enrolment those whose observed trait exceeds a specified value or cutpoint. For patients judged to have extreme values for the screening trait, one might institute therapy whose goal is to bring the extreme values more in line with the norm. Subsequent observation of the trait among those patients entered in the trial will on average tend to be closer to the centre of the distribution compared with initial observation even in the absence of a treatment effect.

Let $\mathbf{Y} = (Y_1, Y_2)$ denote a random vector having a bivariate normal distribution. Denote the means, variances and correlation for Y_1 and Y_2 as $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ and ρ, respectively. Consider the truncated distribution obtained by conditioning on the event $\{Y_1 > a\}$. For any truncation event

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$T = \{Y > a\}$, define

$$\alpha = \frac{a - E(Y)}{\sqrt{V(Y)}},$$

$$C(a) = \frac{\phi(\alpha)}{Q(\alpha)},$$

what is this in words?

$$\phi(z) = \frac{1}{\sqrt{(2\pi)}} \exp(-\tfrac{1}{2}z^2),$$

$$Q(z) = \int_z^\infty \phi(u)\mathrm{d}u. \tag{1}$$

We can easily show that

I should show that

$$E(Y_1 \mid Y_1 > a) = \mu_1 + \sigma_1 C(a),$$
$$E(Y_2 \mid Y_1 > a) = \mu_2 + \rho\sigma_2 C(a).$$

Hence we have

$$E(Y_1 - Y_2 \mid Y_1 > a) = \mu_1 - \mu_2 + (\sigma_1 - \rho\sigma_2)C(a).$$

The quantity

What is this R?

$$R(Y_2 \mid Y_1 > a) = (\sigma_1 - \rho\sigma_2)C(a) \tag{2}$$

is the effect of regression to the mean. In the usual presentation we take Y_1 as the first and Y_2 as the second observation on the same random variable, where $\sigma_1 = \sigma_2 = \sigma$. The above formulation is slightly more general and has some interesting applications. Ederer,[4] James,[5] Gardner and Heady,[6] Cutter,[7] Davis,[8,9] Das and Mulder[2] and Senn[3] discuss regression to the mean and present a variety of practical applications.

Gardner and Heady[6] considered the model

$$Y_i = U + e_i, \tag{3}$$

where Y_i is the ith measurement, U is the unobservable 'true value' of the trait under study, which is fixed for a given subject, and e_i is the error associated with the ith replicate measurement of the same characteristic, $i = 1, \ldots, n$. They assumed that U is normally distributed with mean μ and variance σ_u^2, e is normally distributed with mean 0 and variance σ_e^2, U and e are independent, and e_i and e_j are independent for $i \neq j$. This model corresponds to the situation where the only variability within a subject is that due to independent measurement errors. Under this model,

$$V(Y_i) = \sigma_u^2 + \sigma_e^2,$$

$$\operatorname{corr}(Y_i, U) = \frac{\sigma_u}{\sqrt{(\sigma_u^2 + \sigma_e^2)}},$$

$$\operatorname{corr}(Y_1, Y_2) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2}.$$

With these results, we can show that

$$R(Y_2 \mid Y_1 > a) = R(U \mid Y_1 > a) = \frac{\sigma_e^2}{\sqrt{(\sigma_u^2 + \sigma_e^2)}} C(a). \tag{4}$$

Under this model, if we observe n replicate measurements on a given subject for classification and condition on the event $\{\bar{Y}_n > a\}$, where $\bar{Y}_n$ is the sample mean of the n replicates, then

$$V(\bar{Y}_n) = \sigma_u^2 + (\sigma_e^2/n),$$

and the regression effect on the $(n + 1)$th measurement is

$$R(Y_{n+1} \mid \bar{Y}_n > a) = \frac{\sigma_e^2/n}{\sqrt{[\sigma_u^2 + (\sigma_e^2/n)]}}\, C(a),$$

and it approaches zero as $n \to \infty$. In other words, we can reduce the regression to the mean effect on a given subject by classifying the subject on the basis of a large number of replicate measurements.

STATISTICAL MODEL FOR WITHIN-SUBJECT EFFECT

The model proposed by Gardner and Heady[6] assumes that within a given subject the only variability among repeated observations is that attributable to independent measurement errors. In practice, this assumption is not always reasonable. For example, suppose we want to study some treatment's effect on blood pressure. An individual's blood pressure fluctuates constantly and may vary substantially over a short period. Many factors can influence this within-subject variability, perhaps most notably the subject's emotional state at the time of measurement. If Y_1 and Y_2 are two measurements on a subject, then a more appropriate model might be

$$Y_i = U + S_i + e_i, \quad i = 1, 2, \tag{5}$$

where S_i is the 'subject effect' which may result from biological variability. Note that S_1 and S_2 are not necessarily independent. Whereas we may reasonably use model (3) where within-subject variability is attributable solely to independent measurement errors when we take replicate measurements under identical conditions, measurements taken at different times under different conditions will lead to within-subject fluctuations (for example, fluctuations due to biological variability) and hence model (5) will be more appropriate. As another example, let Y_1 denote some characteristic measured on parents and Y_2 the same characteristic measured on one of the offspring. In this case the subject effects, as defined in (5), may actually be correlated genetic effects but the correlation will be less than unity even in the absence of measurement errors, and will lead to within-subject variability.

Consider the model in (5). In addition to the assumptions made on the model given in (3), assume that $S_i \sim N(0, \sigma_s^2)$, U and S_i are independent, e_i and S_i are independent, and $\operatorname{corr}(S_i, S_j) = \rho_s\ (> 0)$ for $i \neq j$. Then we can show that

$$\boxed{\begin{aligned} \operatorname{corr}(Y_1, Y_2) &= \frac{\sigma_u^2 + \rho_s \sigma_s^2}{\sigma_u^2 + \sigma_s^2 + \sigma_e^2}, \\ R(Y_2 \mid Y_1 > a) &= \frac{(1 - \rho_s)\sigma_s^2 + \sigma_e^2}{\sqrt{(\sigma_u^2 + \sigma_s^2 + \sigma_e^2)}}\, C(a). \end{aligned}} \tag{6}$$

Under model (5), suppose we take, for each subject in the study, repeated measurements at m different times, and at each time we make n replicate measurements; that is, we have

$$Y_{ij} = U + S_i + e_{ij}, \quad i = 1, \ldots, m, \quad j = 1, \ldots, n, \tag{7}$$

where

$$U \sim \mathrm{N}(\mu, \sigma_u^2),$$

$$\mathbf{S} = (S_1, \ldots, S_m) \sim \mathrm{N}(\mathbf{0}, \Sigma),$$

$$\Sigma = \begin{bmatrix} \sigma_s^2 & \rho_s\sigma_s^2 & \cdots & \rho_s\sigma_s^2 \\ & \sigma_s^2 & \cdots & \rho_s\sigma_s^2 \\ & & & \vdots \\ & & & \sigma_s^2 \end{bmatrix}$$

and $e_{ij} \sim$ iid $\mathrm{N}(0, \sigma_e^2)$ and is independent of U and $\mathbf{S}$. Now suppose we use

$$\bar{Y} = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} Y_{ij}$$

to classify subjects; that is, we include subjects in the study and administer treatment only when $\bar{Y}$ exceeds a cutoff point a. We can show that

$$V(\bar{Y}) = \sigma_{\bar{y}}^2 = \sigma_u^2 + (\sigma_s^2/m)[1 + (m-1)\rho_s] + (\sigma_e^2/mn). \tag{8}$$

Now, if we take another observation, say Y^*, after administration of treatment, then

$$\mathrm{corr}(\bar{Y}, Y^*) = \frac{\sigma_u^2 + \rho_s\sigma_s^2}{\{\sigma_u^2 + (\sigma_s^2/m)[1 + (m-1)\rho_s] + (\sigma_e^2/mn)\}^{1/2}(\sigma_u^2 + \sigma_s^2 + \sigma_e^2)^{1/2}}$$

and the effect of regression to the mean simplifies to

$$R(Y^* \mid \bar{Y} > a) = E(\bar{Y} - Y^* \mid \bar{Y} > a) = \frac{(\sigma_s^2/m)(1 - \rho_s) + (\sigma_e^2/mn)}{\{\sigma_u^2 + (\sigma_s^2/m)[1 + (m-1)\rho_s] + (\sigma_e^2/mn)\}^{1/2}} C(a). \tag{9}$$

We can rewrite the regression effect in (9) as

$$R(Y^* \mid \bar{Y} > a) = R_s(Y^* \mid \bar{Y} > a) + R_e(Y^* \mid \bar{Y} > a), \tag{10}$$

where

$$R_s(Y^* \mid \bar{Y} > a) = \{(1 - \rho_s)\sigma_s^2/\sigma_{\bar{y}}\}C(a), \tag{11}$$

$$R_e(Y^* \mid \bar{Y} > a) = \{(\sigma_e^2/mn)/\sigma_{\bar{y}}\}C(a), \tag{12}$$

and $\sigma_{\bar{y}}$ is as defined in (8). In other words, we can decompose the regression to the mean effect R into two components, R_s and R_e, where R_s is the portion attributable to the within-subject variation, and R_e is the portion attributable to the variation due to measurement errors, provided that subject effect and measurement error are independent. Assuming that within-subject variability and measurements errors are the only sources of regression to the mean effects, if we know any two of R_s, R_e and R then we can determine the third. Thus we can sometimes estimate R_s even though we may be unsure of the correlation structure among subject effects.

We can reduce the regression effect due to the independent measurement errors either by increasing the number of repeated measurements m, and/or by increasing the number of replications n of each measurement. On the other hand, we can reduce the regression effect due to within-subject variability only by increasing the number of measurements at different times. We may not find it practical, however, in terms of time and cost to take a large number of measurements before administration of the treatment merely to classify subjects. On the other hand, we may find it reasonable to replicate the measurement at a given time so as to reduce the regression effect attributable to measurement errors. A more practical alternative is to take n

replicate measurements on the characteristic before treatment to obtain a mean (say $\bar{Y}_1$) for classification, and then measure the characteristic again after treatment (say Y_2). As we see from the following discussion, we cannot reduce the effect of within-subject variability under this plan.

In particular,

$$V(\bar{Y}_1) = \sigma_u^2 + \sigma_s^2 + (\sigma_e^2/n)$$

and the regression effect reduces to

$$R(Y_2 \mid \bar{Y}_1 > a) = \frac{\sigma_s^2(1-\rho_s) + (\sigma_e^2/n)}{\sqrt{[\sigma_u^2 + \sigma_s^2 + (\sigma_e^2/n)]}}\, C(a). \tag{13}$$

Taking the limit of (13) as n approaches infinity, we get

$$\lim_{n\to\infty} R(Y_2 \mid \bar{Y}_1 > a) = \frac{\sigma_s^2(1-\rho_s)}{\sqrt{(\sigma_u^2 + \sigma_s^2)}}\, C(a). \tag{14}$$

As we can see from (14), for large n the regression effect approaches 0 as $\rho_s \to 1$, and it is maximum when $\rho_s = 0$. In other words, if $\rho_s < 1$, it is impossible to eliminate the regression effect completely by taking replicate measurements, because of the within-subject variability. ?

To illustrate numerically the behaviour of regression effects for different values of the parameters, we considered model (7) for the following combinations of parameters: $\mu = 100$, $\sigma_u^2 = 100$, $a = 116{\cdot}45$, $\rho = 0{\cdot}1$, $0{\cdot}3$ and $0{\cdot}5$, $(\sigma_s, \sigma_u) = (8, 6)$, $(4, 6)$ and $(4, 3)$, and $(m, n) = (1, 1)$, $(1, 9)$ and $(4, 9)$. The cutoff point 116·45 corresponds to the 95th percentile for the normal distribution N(100, 100). In the absence of within-subject variability and measurement errors, we expect 5 per cent of this distribution to exceed 116·45. In the presence of within-subject variability and measurement errors, $Q(\alpha)$, as defined in (1), is the probability of exceeding 116·45. We call any excess $\{Q(\alpha) - 0{\cdot}05\}$ the probability of misclassification, denoted P_m. A value of $P_m = 0{\cdot}05$ indicates, in this example, that as many would be misclassified as correctly classified. We computed P_m together with the expected regression to the mean R, the amount attributable to within-subject variability R_s, and the amount attributable to measurement error R_e for the combination of parameters given above. The results are presented in Table I. We can see several trends by observing the change in P_m, R, R_s and R_e as we change each of ρ_s, σ_s, σ_e, m and n while holding the other parameters constant:

(i) As the correlation among within-subject effects increases, P_m increases and R, R_s and R_e decrease.
(ii) R_s decreases as σ_s decreases and R_e decreases as σ_e decreases.
(iii) An increase in m decreases both R_s and R_e.
(iv) An increase in n decreases R_e but does not decrease R_s.
(v) Depending on the magnitude of σ_s and σ_e, the regression effect may be substantial for small m and n and may be a nuisance even for m and n of modest magnitude.

We assumed a uniform correlation pattern as is often used in repeated measures analyses for a within-subject correlation structure. In many applications, the correlation may decrease as the time interval separating the measurements increases. One such structure is the autocorrelation model, in which the covariance among subject effects is

$$\Sigma = \begin{bmatrix} \sigma_s^2 & \rho_s\sigma_s^2 & \rho_s^2\sigma_s^2 & \cdots & \rho_s^m\sigma_s^2 \\ & \sigma_s^2 & \rho_s\sigma_s^2 & \cdots & \rho_s^{m-1}\sigma_s^2 \\ & & & & \vdots \\ & & & & \sigma_s^2 \end{bmatrix}.$$

Table I. Probability of misclassification P_m and regression to the mean effects (R, R_s and R_e) for model (7) with $\mu = 100$, $\sigma_u = 10$ and $a = 116{\cdot}45$

rho close to 0 => high R

repeated measurements decrease R a lot

smaller variability => smaller R

ρ_s	σ_s	σ_e	m	n	P_m	R	R_s	R_e
0·1	8	6	1	1	0·072	10·97	6·75	4·22
			1	9	0·052	8·29	7·75	0·54
			4	9	0·018	2·69	2·52	0·17
	4	6	1	1	0·041	7·35	2·10	5·25
			1	9	0·017	3·26	2·55	0·71
			4	9	0·005	0·90	0·70	0·20
	4	3	1	1	0·021	3·73	2·19	1·54
			1	9	0·014	2·50	2·32	0·18
			4	9	0·006	0·67	0·62	0·05
0.3	8	6	1	1	0·072	9·47	5·25	4·22
			1	9	0·052	6·57	6·03	0·54
			4	9	0·026	2·00	1·84	0·16
	4	6	1	1	0·041	6·89	1·63	5·25
			1	9	0·017	2·69	1·98	0·71
			4	9	0·007	0·73	0·54	0·19
	4	3	1	1	0·021	3·46	1·92	1·54
			1	9	0·014	2·21	2·03	0·18
			4	9	0·007	0·59	0·54	0·05
0·5	8	6	1	1	0·072	7·97	3·75	4·22
			1	9	0·052	4·85	4·31	0·54
			4	9	0·033	1·40	1·24	0·16
	4	6	1	1	0·041	6·42	1·17	5·25
			1	9	0·017	2·12	1·42	0·70
			4	9	0·009	0·57	0·38	0·19
	4	3	1	1	0·021	2·91	1·37	1·54
			1	9	0·014	1·63	1·45	0·18
			4	9	0·009	0·43	0·38	0·05

We can derive regression to the mean effects for this and other covariance structures without difficulty.

EXAMPLE: REGRESSION EFFECTS IN A CHOLESTEROL STUDY

A study involved serum cholesterol measurements on 107 medical students collected during their freshman and senior years; the means were 185 mg/100 ml and 176 mg/100 ml, respectively. The standard deviation was 31 mg/100 ml for each of the two groups, and the correlation between the two groups was 0·53. The data ranged from 93–250 mg/100 ml during the freshman year and from 95–263 mg/100 ml during the senior year. Although one often employs a log-normal distribution for serum cholesterol levels, we found the normal distribution satisfactory for these data.

Thirty freshman students had cholesterol levels that exceeded 200; for these 30 students the mean was 223 mg/100 ml with a standard deviation of 14 mg/100 ml. In their senior year these 30 students had a mean level of 199 mg/100 ml with a standard deviation of 29 mg/100 ml. Thus the observed decrease was 24 mg/100 ml. Substituting the estimates based on the data for all 107 students for the actual parameters, we have $E(Y_1) = 185$, $E(Y_2) = 176$, $\sigma_1 = \sigma_2 = 31$, $\rho = 0{\cdot}53$ and $a = 200$. We used these parameter estimates in equation (1) to get $\alpha = 0{\cdot}484$, $\phi(\alpha) = 0{\cdot}355$,

$Q(\alpha) = 0{\cdot}361$ and $C(a) = 0{\cdot}982$. The decrease expected as a result of regression to the mean is thus, from equation (2),

$$R(Y_2 \mid Y_1 > 200) = 14{\cdot}3 \text{ mg/100 ml.}$$

Since the difference in means for the entire group of 107 students was 9 mg/100 ml, the estimated decrease for the 30 students whose levels exceeded 200 mg/100 ml in their freshman year was $9 + 14{\cdot}3 = 23{\cdot}3$ mg/100 ml. This is in line with the decrease of 24 mg/100 ml actually observed.

Now, the correlation between freshman and senior year cholesterol levels is obviously attenuated. From blind duplicate determinations, an estimate of the component of variability for measurement error was $\hat{\sigma}_e^2 = 49$ (mg/100 ml)^2. Since the variance of serum cholesterol levels over the sample was $(31)^2 = 961$ (mg/100 ml)^2, if the only source of attenuation was measurement errors we would expect a correlation of $(961 - 49)/961 = 0{\cdot}95$ between the two groups. This is considerably higher than the observed correlation 0·53, and therefore we suspect a substantial within-subject variability. Since the blind duplicates were taken on an external sample and the observed decrease was based on single measurements during freshman and senior years, we have $m = n = 1$. From equation (12) we get the regression effect attributable to measurement error $R_e = 1{\cdot}5$ mg/100 ml, and using $R_s = R - R_e$ we obtain 12·8 mg/100 ml as an estimate of the within-subject regression effect. Although no specific treatment was given to students with high freshman cholesterol levels, it is likely that some sought treatment on their own, and so 12·8 mg/100 ml probably overestimates R_s in this example. Davis[8] reported 0·82 as an estimate of the correlation between cholesterol levels measured on two different occasions without treatment intervention for 1346 men 35–59 years of age.

DISCUSSION

Any extraneous source of variability that has an attenuating effect on the correlation between two random variables may contribute to regression to the mean effects. Two important sources are the within-subject effects and measurement errors. Gardner and Heady[6] demonstrated the effect of measurement error in the absence of any other source of within-subject variability. Within-subject effects may be absent in many applications, and similarly in many situations measurement is so precise that measurement errors may be negligible. On the other hand, there are many applications where both within-subject variability and measurements errors can have a substantial effect on regression to the mean, and hence on the conclusions and interpretations of the analysis.

Investigators should exercise caution with studies where eligibility depends on a quantitative characteristic judged as too high or too low in relation to some norm. If one wishes to include in the study a certain percentage of the true distribution for the characteristic, then within-subject variability and measurement errors can lead to both a high misclassification rate (false positives) and a substantial regression to the mean effect on subsequent observation. We can reduce measurement error by such steps as improving instrumentation, making observations under identical conditions, sharpening definitions used in the measuring process and training observers to measure with precision. It is also feasible in some instances to control within-subject variability. In a study of hypertension, for example, we can make sure that blood pressures are measured under controlled conditions, paying special attention to measuring the pressure when the subject is relaxed. Serum cholesterol levels can be measured after the subject has fasted for 12 hours. We can monitor conditions to make sure that strict adherence to protocol is maintained each time a subject is observed. We can 'average out' within-subject effects and measurement error effects by taking both repeated and replicated measurements during the process of screening patients for study. This strategy may be feasible in large, multicentre trials that entail long-term follow-up, but it can be costly and time-consuming with small short-term trials and other such studies. Ederer[4] proposed taking a single measurement for classification purposes, a second measurement as a baseline and then a third post-treatment measurement. This plan may be efficient in many applications. Even if these steps are taken, the importance of using a control group in designed studies must not be overlooked both for the usual reasons and to enhance our ability to make good estimates of treatment effects in the presence of regression to the mean (see, for example, Davis[8]).

In the absence of external estimates of the parameters required to estimate regression to the mean, James[5] gives a method of moments estimator of the regression effect. Senn and Brown[10] discuss the method of moment estimator and argue that the maximum likelihood estimator is preferable, although the latter must be obtained by numerical iteration. Mee and Chua[11] describe a *t*-test for comparing the means of paired data in the presence of regression to the mean.

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